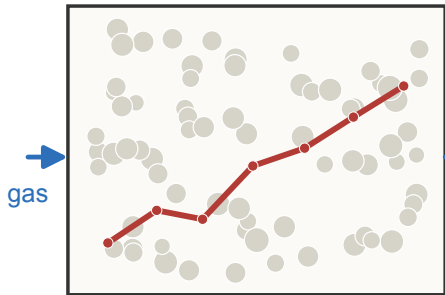


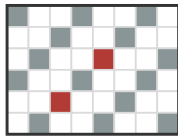
# Separating fines breakthrough from pore-network clogging indicators

## DEM transport

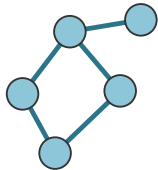


fines migrate through a stiff pebble bed

## Pore-structure observables

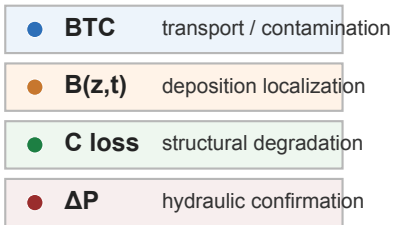


local void/  
solid map



connected-pore  
skeleton

## Staged assessment



breakthrough alone is not a  
pressure-calibrated clogging criterion

DEM trajectories, pore reconstruction and bounded pressure checks provide complementary indicators in the sampled cases.